

1. PURPOSE AND SCOPE

This runbook defines Standard Operating Procedures (SOPs) for the Network Operations Center (NOC) managing MPLS and SD-WAN infrastructure. It covers fault isolation, escalation paths, and restoration procedures for Tier-1 through Tier-3 incidents.

2. BGP TROUBLESHOOTING PROCEDURE

BGP Route Flapping:

Symptom: Routes appear and disappear from the routing table repeatedly.

Step 1: Run 'show bgp summary' to identify unstable neighbors.

Step 2: Check interface counters - 'show interface <if>' for CRC errors.

Step 3: Enable route flap dampening: 'bgp dampening 15 750 2000 60'.

Step 4: If flapping persists, isolate the peer with 'neighbor <ip> shutdown'.

Step 5: Escalate to Tier-2 if not resolved within 30 minutes.

BGP Neighbor Down:

Step 1: Verify physical connectivity - ping the neighbor IP.

Step 2: Check BGP timers: 'show bgp neighbors <ip> | inc hold'.

Step 3: Verify AS number matches: 'show run | section router bgp'.

Step 4: Clear BGP session: 'clear ip bgp <ip> soft'.

Step 5: If MD5 authentication mismatch, re-enter password on both peers.

3. MPLS LINK FAILURE PROCEDURE

Symptom: Traffic loss on MPLS circuit, LDP session dropped.

Step 1: Confirm alarm in NMS - identify affected PE router and circuit ID.

Step 2: Check LDP session: 'show mpls ldp neighbor'.

Step 3: Verify RSVP-TE tunnels: 'show mpls traffic-eng tunnels brief'.

Step 4: Test LSP ping: 'ping mpls ipv4 <prefix>/32 repeat 100'.

Step 5: If physical failure confirmed, open ticket with carrier.

Step 6: Activate backup path: 'no shutdown' on backup interface.

Step 7: Verify traffic restored via NetFlow dashboard within 5 minutes.

Escalation: P1 incidents must be escalated to Tier-2 within 15 minutes.

4. SD-WAN OVERLAY TROUBLESHOOTING

Symptom: Branch site loses connectivity to data center.

Step 1: Log into vManage - navigate to Monitor > Network.

Step 2: Check tunnel status: Control Connections must show 'Up'.

Step 3: Verify BFD sessions: 'show sdwan bfd sessions'.

Step 4: Check policy application: 'show sdwan policy from-vsmart'.

Step 5: Restart WAN Edge if no control connections: 'request system reboot'.

Step 6: If vSmart unreachable, failover to secondary controller.

Critical: Never reboot more than one WAN Edge in the same region simultaneously.

5. ESCALATION MATRIX

P1 (Critical) - Service Down : Escalate to Tier-2 in 15 min, NOC Manager in 30 min.

P2 (High) - Degraded Service : Escalate to Tier-2 in 1 hour.

P3 (Medium) - Minor Impact : Resolve within 4 hours or escalate.

P4 (Low) - No Impact : Resolve within 24 hours.

Escalation contacts:

Tier-2 Network Engineer : +91-11-2345-6789 (24x7)

NOC Manager : +91-11-2345-6790

Vendor TAC (Cisco) : 1-800-553-2447

ISP NOC (BSNL Leased) : 1800-180-1503

6. CHANGE MANAGEMENT GUIDELINES

All changes to production routers must follow this process:

1. Raise Change Request (CR) in ServiceNow at least 48 hours before.
 2. Obtain approval from Network Architect and NOC Manager.
 3. Schedule during approved maintenance window: Saturday 02:00-05:00 IST.
 4. Have rollback plan documented before starting.
 5. Test in staging environment if available.
 6. Post-change verification: ping, traceroute, BGP summary check.
- Emergency changes during P1 incidents are exempt but must be documented within 2 hours.